

CONCEPT OF GENE

A gene is the basic physical and functional unit of heredity.

Genes are made up of DNA.

Gregor John Mendel stated that **hereditary units**, which he called **factors** (now known as **genes**), are responsible for transferring genetic traits from parents to offspring.

Yet the mystery of the 'hereditary units' was unrevealed during early 1900s.

In 1909, **Watson Johanson** coined the term 'gene' that act as hereditary units.

Sutton and **Boveri** proposed Chromosome theory of Heredity.

In, 1906, **W.Bateson** and **R. C Punnet** reported the first case of linkage in sweet pea and proposed the presence or absence theory

Major Contributions/History

In 1969, **Shapiro and co-workers** published the first picture of isolated genes.

They purified the lac operon of DNA and took photographs through electron microscope.

In 1941, **G.W. Beadle** and **E.L. Tatum** working at St Stanford university clearly demonstrated **onegene-one enzyme hypothesis**, based on experiments on *Neurospora crassa*.

One Gene-One Enzyme Hypothesis-

Beadle and Tatum made it clear that genes are the functional units and transmitted to progenies over generations; also, they undergo mutations.

They treated *N. crassa* with X-rays and selected for X-ray induced mutations that would have been lethal.

Their selection would have been possible when *N. crassa* was allowed to grow on nutrient medium containing vitamin B6.

This explains that X-rays mutated vitamin B6 synthesizing genes.

They concluded that a gene codes for the synthesis of one enzyme.

In 1958, Beadle and Tatum with Lederberg received a Nobel prize for their contribution to physiological genetics.



Figure 1: Diagrammatic Representation of Beadle and Tatum Experiment
Gene Theory

This theory of gene was well accepted by the cytologists.

In 1933, Morgan was awarded Nobel prize for advocating the theory of genes.

After the discovery of DNA as carrier of genetic information, the Morgan's theory was discarded.

Therefore, it is necessary to understand both, the classical and modern concepts of gene.

According to the classical concepts a gene is a unit of

- (i) physiological functions,
- (ii) transmission or segregation of characters, (iii) mutation.

Chemical Nature of Gene

DNA-

Deoxyribonucleic Acid

Double helix

Carries genetic information

Located in the nucleus

The monomer is a nucleotide

— A phosphate

- A ribose sugar
 - A nitrogenous base
- Bases in DNA:**

A – adenine
T – thymine
C – cytosine
G – guanine

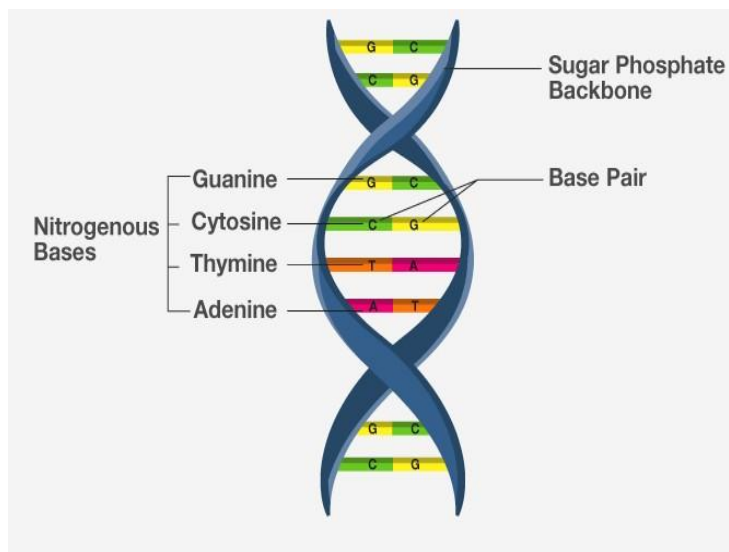


Figure 2: Structure of DNA

Nucleotide It is the units that make up DNA.

Nucleotides are composed of a 5- carbon sugar (deoxyribose), a phosphate group, and a nitrogen base (each nucleotide has only one of the four; A, C, T, G).

Purines- heterocyclic aromatic organic compound, consisting of a pyrimidine ring fused to a imidazole ring.

Pyrimidines- heterocyclic aromatic compound similar to benzene and pyridine, containing 2 nitrogen atoms at positions 1 and 3 of the six-member ring.

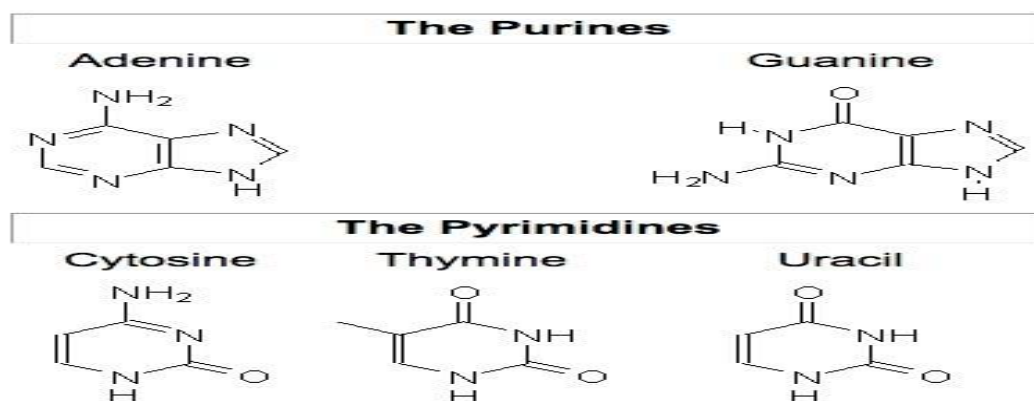


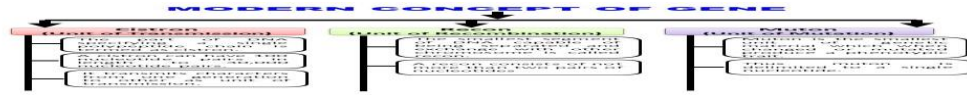
Figure 3: Structure of Purines and Pyrimidines

Classical Concept of Gene

Classical concept of gene was introduced by Sutton (1902) and was elaborated by Morgan (1913). Bidge (1923), Muller (1927) and others which outlined as follows:

- Genes are discrete particles inherited in Mendelian fashion that occupies a definite locus in the chromosome and responsible for expression of specific phenotypic character.
- Numbers of genes in each organism is more than no. of chromosome, hence several genes are located on each chromosome.

- iii. The genes are arranged in a single linear order like beads on string.
- iv. Each gene occupies specific location called **locus**.
- v. If the position of gene changes, then character changes.
- vi. Gene can be transmitted from parents to offsprings. vii. Genes may exist in many alternate forms called **alleles**.
- viii. Genes may undergo sudden changes in position and composition called **mutation**.
Genes are capable of self-duplication producing their own copies.



After the discovery of DNA, the gene has been defined as cistron, recon and muton. The classical gene is the smallest unit that could undergo a mutational change. A gene further divided into smaller units of function, mutation and recombination n.

Symour Benzer (1955-USA) coined the terms cistron, recon, and muton to explain the relationship between DNA and genetic phenomena.

- a) **Cistron:** It is the unit of function. C istron represents a segment of the D N A molecule and consists of a linear sequence of nucleotides, which controls some cellular function. In *E. Coli* cistron may contain about 1500 base pairs. Some cistrons may contain as many as 30,000 base pairs. The cistron begin with initiation codon and ends with a terminating codon. Each cistron is responsible for coding one m-RNA molecule which in turn controls the formation of one polypeptide chain. Each cistron consists of hundreds of mutons and recons.
- b) **Recon:** It is a unit of recombination. It is the smallest unit capable of recombining genetically. Recombination studies on microbes indicate that structurally the recon consists of one or two pairs of nucleotides, possibly only one pair.
- c) **Muton:** It is a unit of mutation. The shortest chromosomal unit capable of undergoing mutation has been called the muton. The muton consists of one or many pairs of nucleotides within the DNA molecule.

Molecular Concept of Gene

In molecular terms, a gene commonly is defined as the entire nucleic acid sequence that is necessary for the synthesis of a functional polypeptide.

According to this definition, a gene includes more than the nucleotides encoding the amino acid sequence of a protein, referred to as the **coding region**.

A gene also includes all the DNA sequences required for synthesis of a particular RNA transcript.

In some prokaryotic genes, DNA sequences controlling the initiation of transcription by RNA polymerase can lie thousands of base pairs from the coding region.

TYPES OF GENES

1. Housekeeping Genes (Constitutive Genes)

These genes are always active because their products are needed for basic life processes of the cell.

Example: genes for glycolysis, ATPase.

2. Non-constitutive Genes (Luxury Genes)

These genes are not always active. They work only when needed by the cell.

Example: nitrate reductase gene in plants, lactose genes in *E. coli*.

They are of two types: inducible and repressible.

3. Inducible Genes

These genes are normally OFF and become ON only in the presence of a specific substance (inducer).

Example: nitrate induces nitrate reductase gene.

4. Repressible Genes

These genes are normally ON but become OFF when the end product is present.

This stopping of activity is called feedback repression.

5. Multigenes (Multiple Gene Family)

A group of similar genes that produce similar products at different times or tissues.

Example: globin gene family.

6. Repeated Genes

Genes present in many copies because their products are needed in large amounts.

Example: rRNA, tRNA, histone genes.

7. Single-Copy Genes

Genes present in only one copy in the genome.

They form 60–70% of functional genes and mainly code for proteins.

8. Pseudogenes

Genes that look like normal genes but do not work due to mutations.

They cannot produce functional proteins.

9. Processed Genes

Genes that lack introns and are formed by reverse transcription.

They are usually non-functional because they lack promoters.

10. Split Genes

Genes that contain coding parts (exons) and non-coding parts (introns).

Introns are removed before RNA leaves the nucleus.

Common in eukaryotes.

11. Transposons (Jumping Genes)

DNA segments that can move from one place to another in the genome.

Discovered by Barbara McClintock in maize.

12. Overlapping Genes

Two genes that share a part of the same DNA sequence.

One DNA segment codes for parts of two different proteins.

13. Structural Genes

Genes that code for proteins or RNAs needed for cell structure and function.

Example: enzymes, hormones, hemoglobin, antibodies, rRNA, tRNA.

14. Regulatory Genes

Genes that control the activity of other genes instead of making proteins.

They include promoters, operators, repressors, and terminators.

15. Tissue-Specific Genes

Genes that are active only in certain tissues.

Example: insulin gene in pancreas, hemoglobin gene in RBCs.

16. Junk Genes (Junk DNA)

DNA sequences that do not code for proteins.

Earlier thought useless, but now known to help in evolution and gene regulation.

Functions of Genes

- 1) Genes are components of genetic material and are thus units of inheritance,
- 2) They control the morphology or phenotype of individuals,
- 3) Replication of genes is essential for cell division,
- 4) Genes carry the hereditary information from one generation to the next,
- 5) They control the structure and metabolism of the body,
- 6) Reshuffling of genes at the time of sexual reproduction produces variations,
- 7) Different linkages are produced due to crossing over,
- 8) Genes undergo mutations and change their expression,
- 9) New genes and consequently new traits develop due to reshuffling of exons and introns.
- 10) Genes change their expression due to position effect and transposons.
- 11) Differentiation or formation of different types of cells, tissues and organs in various parts of the body is controlled by expression of certain genes and non-expression of others,
- 12) Development or production of different stages in the life history is controlled by genes